

LLM-4-RECSYS

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HSE

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진승민 / Джин Сын Мин

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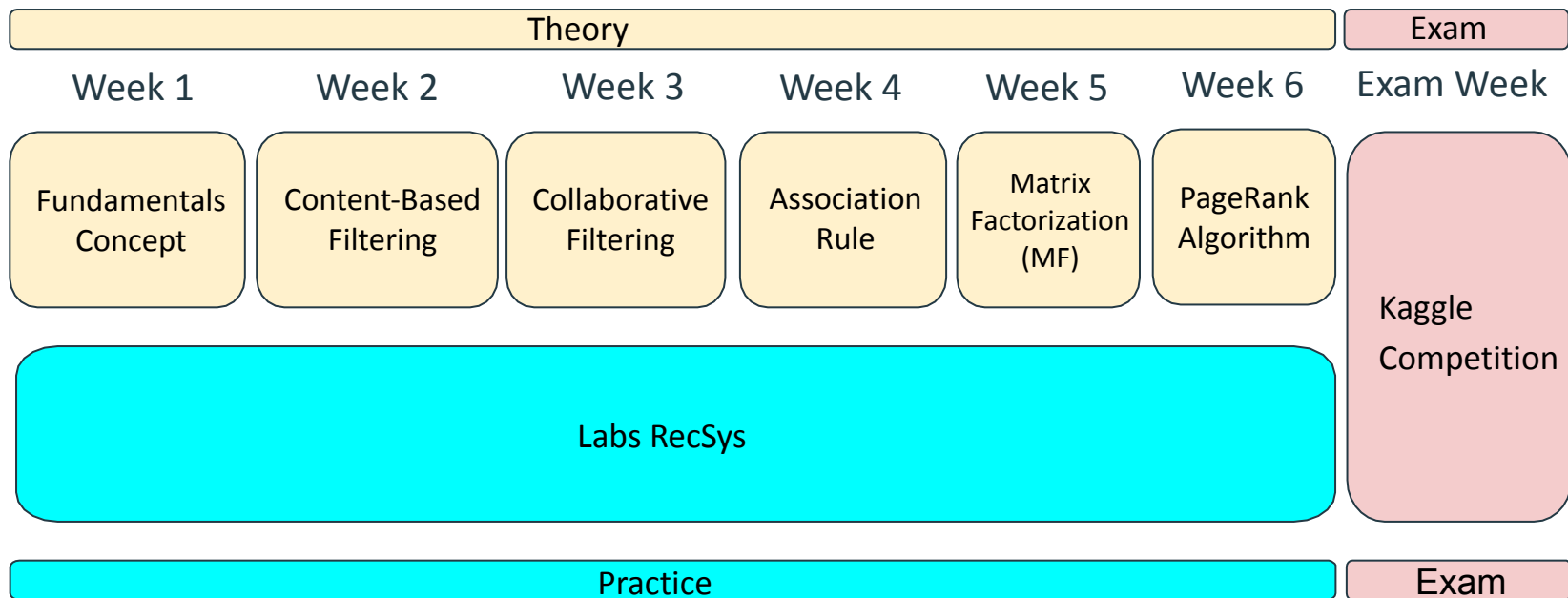
International Laboratory for Intelligent Systems and Structural Analysis
Faculty of Computer Science, HSE University, Moscow, Russia

Mail: sedzhin@hse.ru **LinkedIn:** <https://www.linkedin.com/in/dryjins>

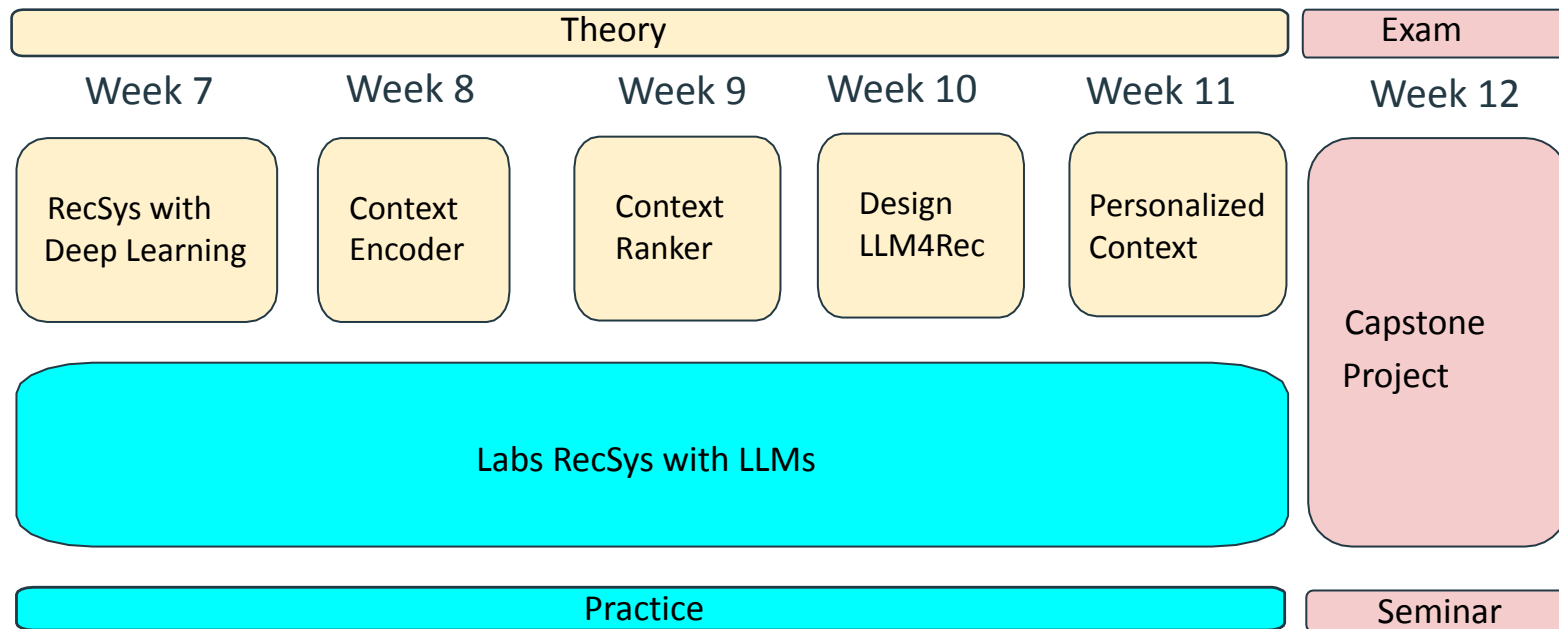
Research Focus : eXplainable AI, Deep Learning, Visual Analytics



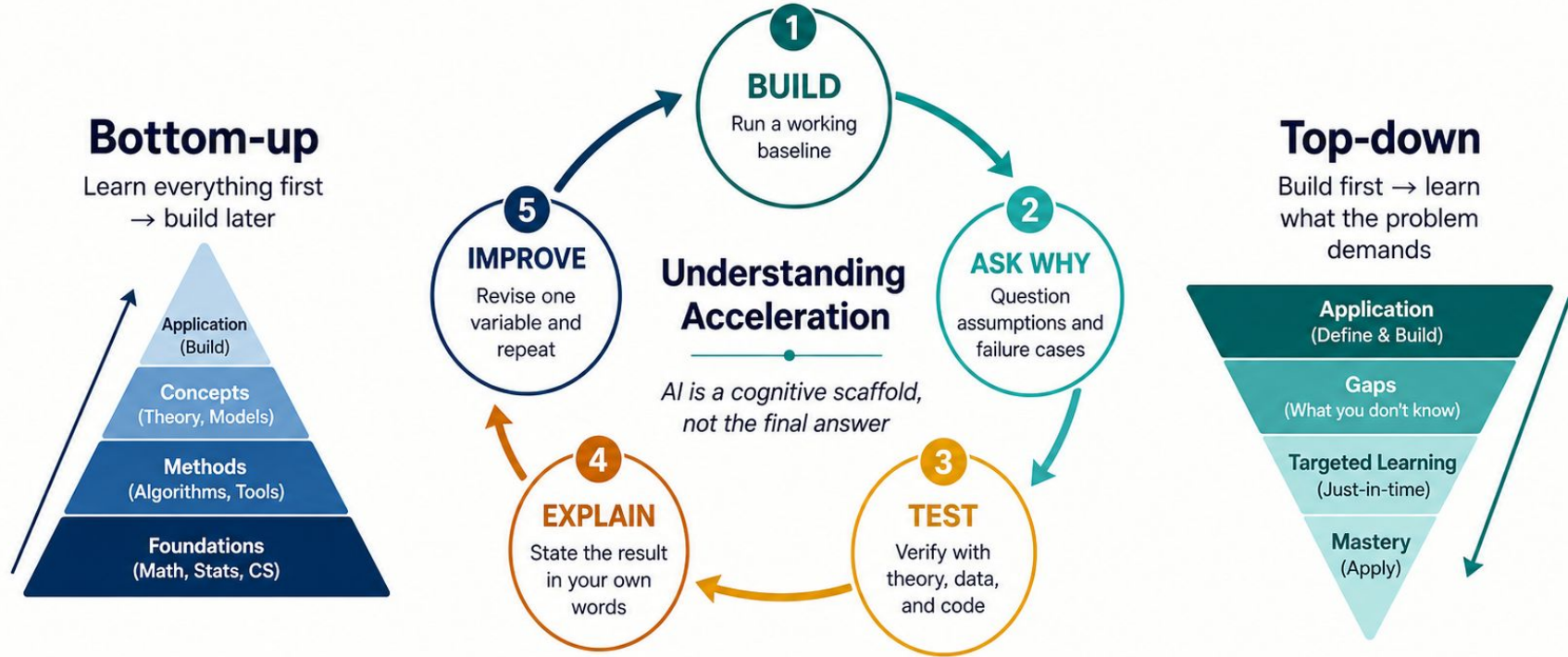
Course Overview Part 1



Course Overview Part 2



AI-Assisted Top-Down Learning



“ AI does not do the learning for you. It shortens the loop through which you learn. ”



2016

AlphaGo changed Go education.

AI-assisted learning



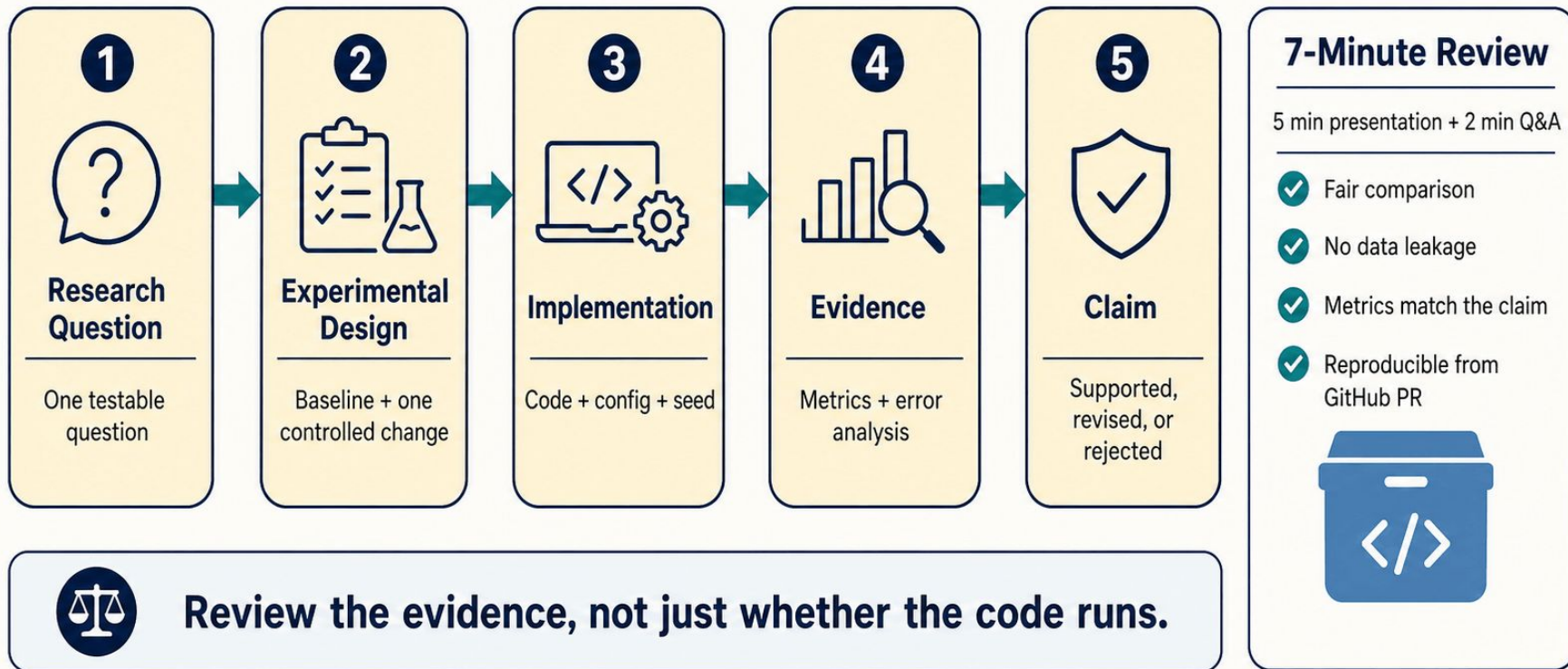
2026

Shin Jin-seo defeated KataGo 2-1

Official two-stone handicap match

AI does not replace expertise. It accelerates the learning loop: generate, verify, critique, reconstruct.

Code Review Session: From Code to Evidence



Midterm Exam

1. Hackathon (24 Hours)
2. Leaderboard Score
3. Jupyter Notebook



≡ kaggle

+ Create

🏠 Home

🏆 Competitions

📁 Datasets

👤 Models

📊 Benchmarks

<> Code

💬 Discussions

🎓 Learn

▼ More

🔍 Search

Sign In

Register

Titanic - Machine Learning from Disaster

Overview Data Code Models Discussion Leaderboard Rules

🔍 Search leaderboard

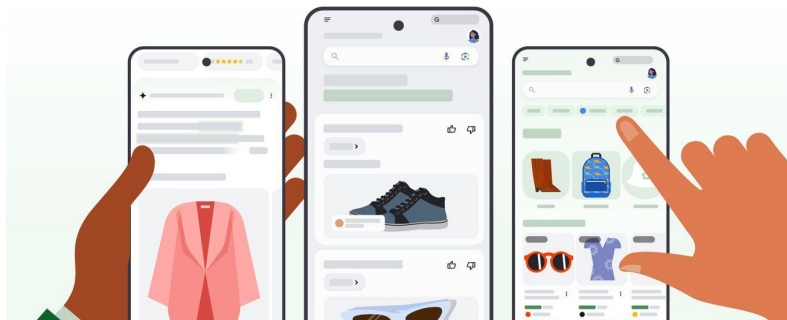
This leaderboard is calculated with all of the test data.

#	Team	Members	Score	Entries	Last	Join
1	hato bai		1.00000	32	1d	
2	Veniamin_Nelin		1.00000	356	4h	
3	Navdhillon72		1.00000	1	2mo	

Example: [CMPE256 - S23 - Book recommendations | Kaggle](#)

Capstone Project

1. 2 - 3 Members
2. Group Presentation (~10 min)
3. Real app required
4. Short paper
5. Fully blocking



Sample IEEE Paper for A4 Page Size

First Author*, Second Author*, Third Author*

*First Third Department, First Third University

Address

Second Country

Address Including Country Name

Abstract. This document gives formatting instructions for authors preparing papers for publication in the Proceedings of an IEEE conference. The authors must follow the instructions given in the document for the papers to be published. You can use this document as both an instruction set and as a template from which you can type your own text.

Keywords.— Include at least 5 keywords on phrases

I. Introduction

This document is a template. An electronic copy can be downloaded from the conference website. For questions on paper guidelines, please contact the conference publications committee as indicated on the conference website. Information about final paper submission is available from the conference website.

II. Fast Layout

An easy way to comply with the conference paper formatting requirements is to use this document as a template and simply type your text into it.

A. Page Layout

Your paper must use a page size corresponding to A4, which is 210mm (8.27") wide and 297mm (11.69") long. The margins must be set as follows:

- Top = 10mm (0.75")
- Bottom = 10mm (0.75")
- Left = Right = 14.2mm (0.56")

Your paper must be in two column format with a space of 4.2mm (0.17") between columns.

B. Font Size

All paragraphs must be indented. All paragraphs must be justified, i.e. both left-justified and right-justified.

C. Text Font of Your Document

The entire document should be in Times New Roman or Times font. Type 3 fonts must not be

used. Other font types may be used if needed for special purposes. Recommended font sizes are shown in Table I.

D. Title and Author Details

Title must be in 14 pt Regular font. Author name must be in 11 pt Regular font. Author affiliation must be in 10 pt font. Email address must be in 9 pt Courier Regular font.

TABLE I
Font Sizes in Paper

Font Size	Appearance (in Times New Roman or Times)	Font	Style
14	title, chapter title, figure caption, reference item (periodic)	Regular	bold
12	author name, abstract body, abstract heading (same as body)	Regular	bold
11	author affiliation, email address, author affiliation	Regular	bold
10	title	Regular	bold

All title and author details must be in single-column format and must be centered.

Every word in a title must be capitalized except for short minor words such as "a", "an", "and", "at", "by", "for", "from", "if", "in", "into", "on", "of", "off", "the", "to", "with".

Author details must not show any professional title (e.g. Managing Director), any academic title (e.g. Dr.) or any membership of any professional organization (e.g. Senior Member IEEE).

To avoid confusion, the family name must be written as the last part of each author name (e.g. John A.K. Smith).

Each affiliation must include, at the very least, the name of the company and the name of the

[IEEE Conference Template - Overleaf, Online LaTeX Editor](#)

Grading

Assessment Component	Weight	Format	Timing	Key Deliverables
Practical Review Session	20%	Individual practical presentation and Q&A	Study Period	GitHub pull request, experimental results, presentation
Midterm Exam: 24-Hour RecSys Kaggle Hackathon	20%	Individual 24-hour hackathon	End of 1st module	Kaggle submission and executable Jupyter Notebook
Final Capstone: LLM4RecSys Short Paper	40%	Team short paper with individual defense	Defense in Week 12	4–6 page short paper, repository, AI review, response-to-review
Class Activity	10%	Individual participation and peer feedback	Study Period	Questions, peer reviews, discussions, and practical-session participation
Attendance	10%		Study Period	
Total	100%			

Please note: Your score is calculated by **dividing the rounded total score by 10**. The **final grade distribution** is strictly capped at **30% (10, 9, 8)**, **40% (7, 6)**, and **30% (5, 4, or Fail) [$\pm 5\%$]**.

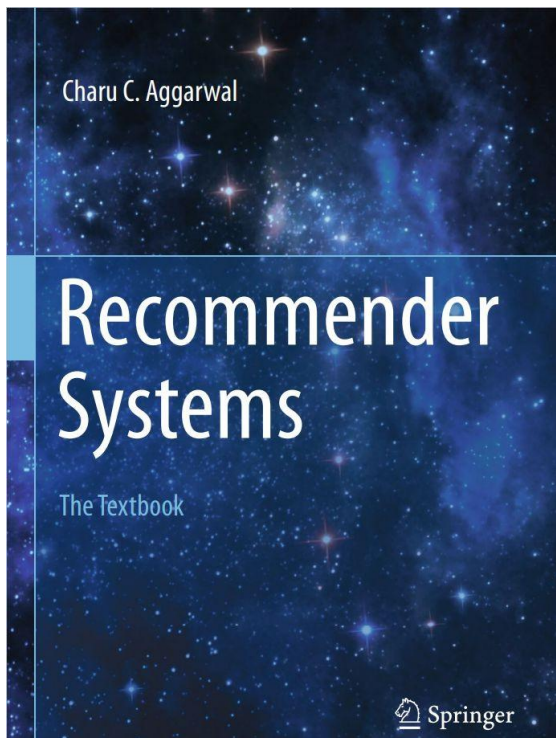
Attendance

Attendance Rate (%)	Score (Points)
Less than 50%	0
50% to less than 70%	1
71% – 73%	2
74% – 76%	3
77% – 80%	4
81% – 83%	5
84% – 86%	6
87% – 90%	7
91% – 93%	8
94% – 96%	9
97% – 100%	10

Notes

- Attendance is defined as full participation from start to finish in scheduled lectures, seminars, and practical classes.
- If you need explicit handling for borderline values (such as values between 70% and 71%), please let me know.

Text book



Recommender Systems: The Textbook, Aggarwal)

RecSys 2023 Tutorial: Large Language Models for Recommendation



Wenyue Hua¹,

Lei Li²,

Shuyuan Xu¹,

Li Chen²

Yongfeng Zhang¹

¹Rutgers University, ²Hong Kong Baptist University

RecSys 2023 Tutorial: Large Language Models for Recommendation

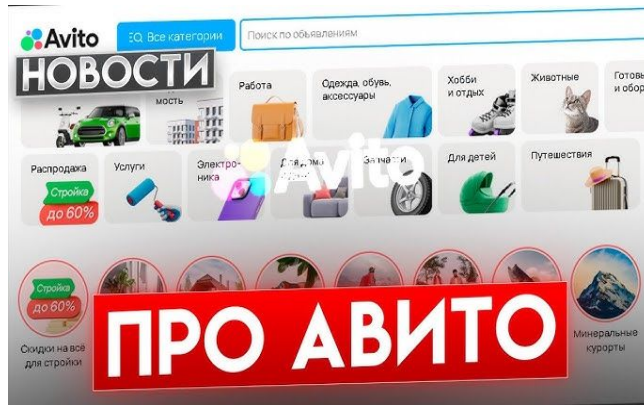


Understanding is the new bottleneck

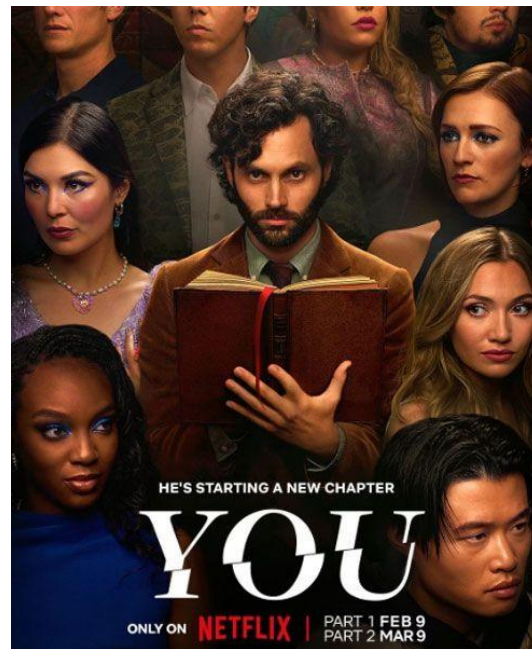
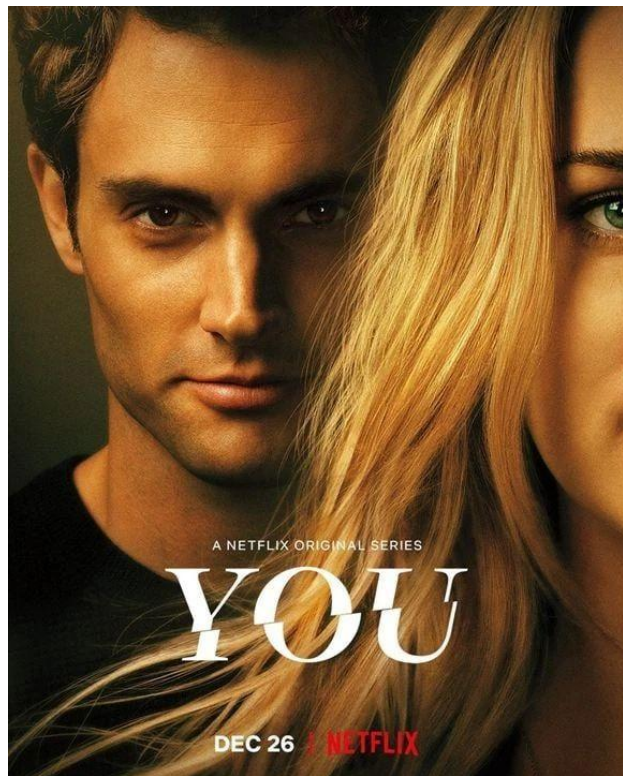
What is RecSys?

Personalised
Experience
(Preference)
Recommendation

Market Demand



What is RecSys?



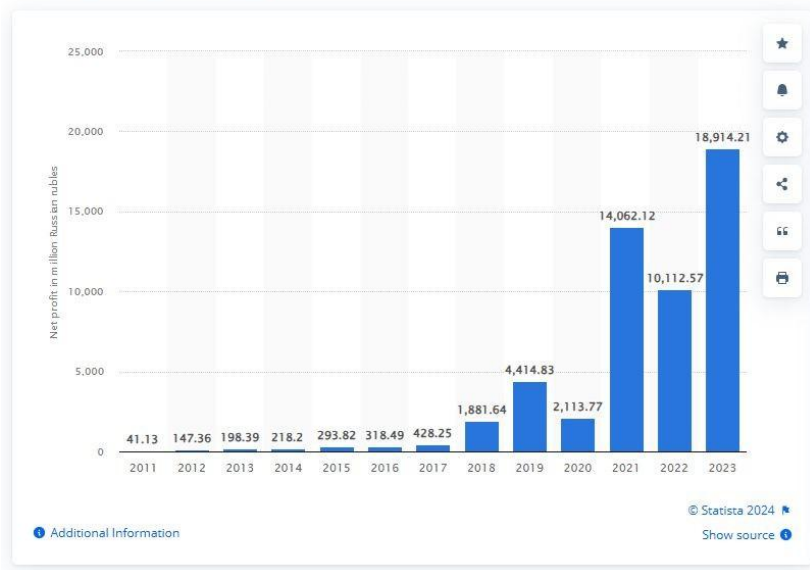
More **You Know**
More **You Like**

Why do we care RecSys?

Personalized Experience Matters in Business Profit.

E-Commerce > B2C E-Commerce

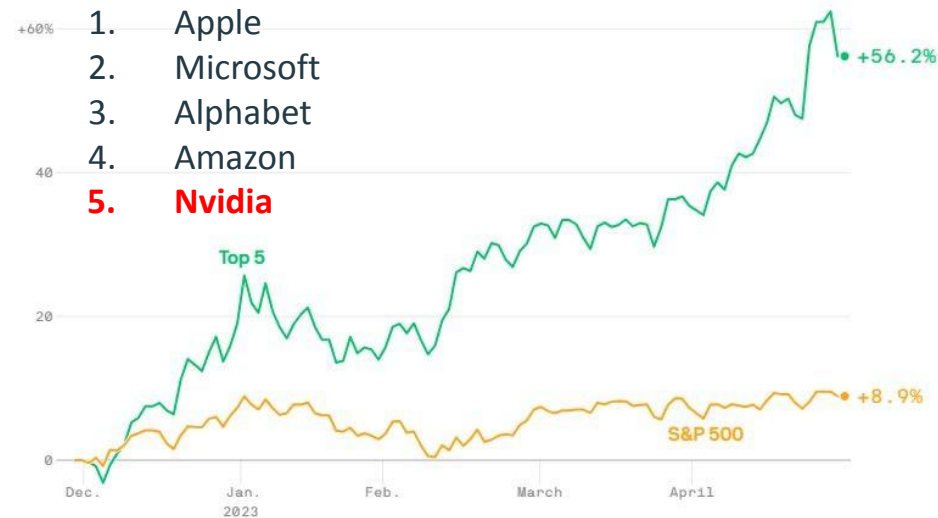
Annual net profit of Wildberries in Russia from 2011 to 2023
(in million Russian rubles)



[Russia: Wildberries net profit 2023 | Statista](#)

Average return of five largest S&P 500 companies

Year-to-date performance; Daily; Dec. 31, 2022, to May 31, 2023



Data: FactSet; Chart: Axios Visuals

[These five companies are responsible for S&P 500's 2023 gains \(axios.com\)](#)

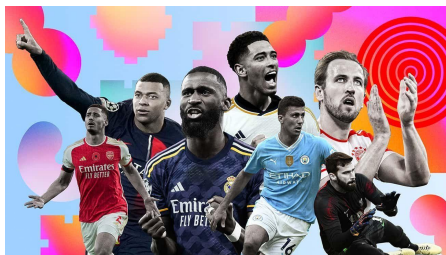
Content-Based Filtering



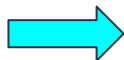
1. Item attributes
2. Similarity
3. Profile matching
4. Personalization

Personal Shopper who knows your history very well!

Collaborative Filtering



Users



Getting Recommendation from close friends!

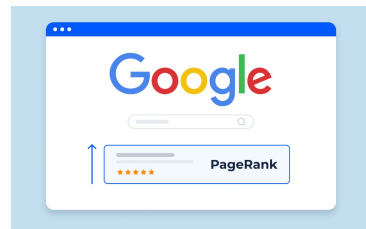
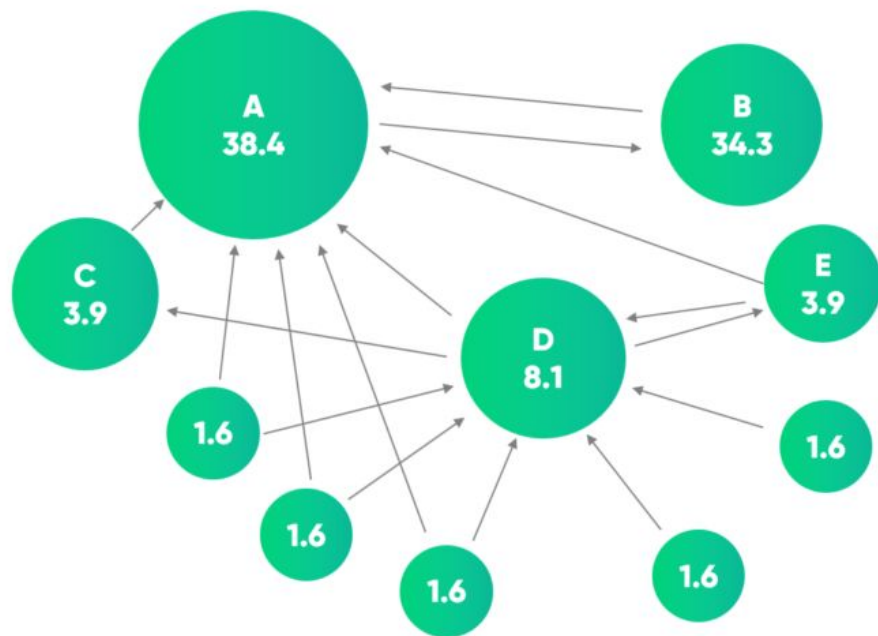


Item



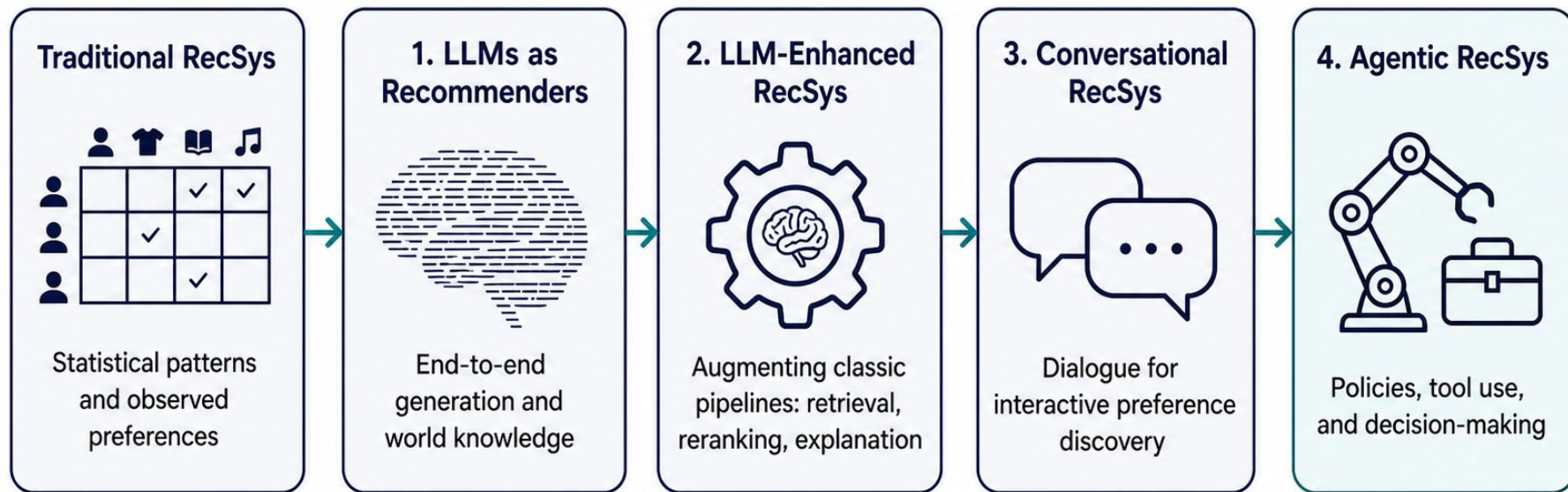
Page Rank Algorithm

Page Rank



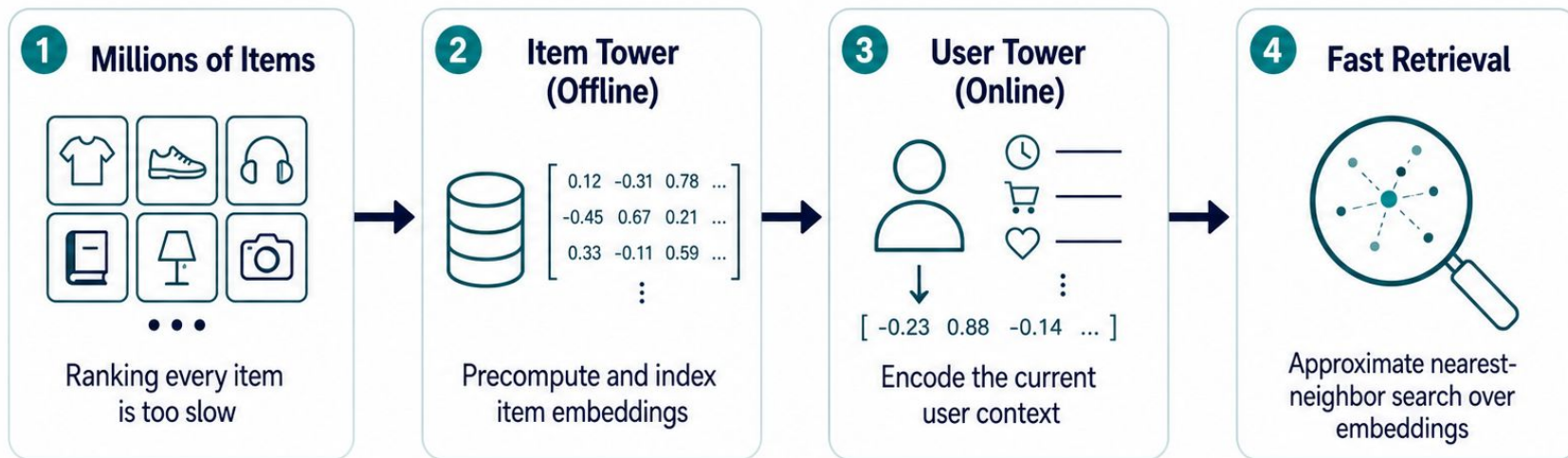
RecSys under
the influence

The Evolution of Recommender Systems



From preference prediction to context-aware interaction and action.

Why Two-Tower Retrieval? Industrial-Scale Speed



$$\text{score}(\text{user}, \text{item}) = \text{user embedding} \cdot \text{item embedding}$$

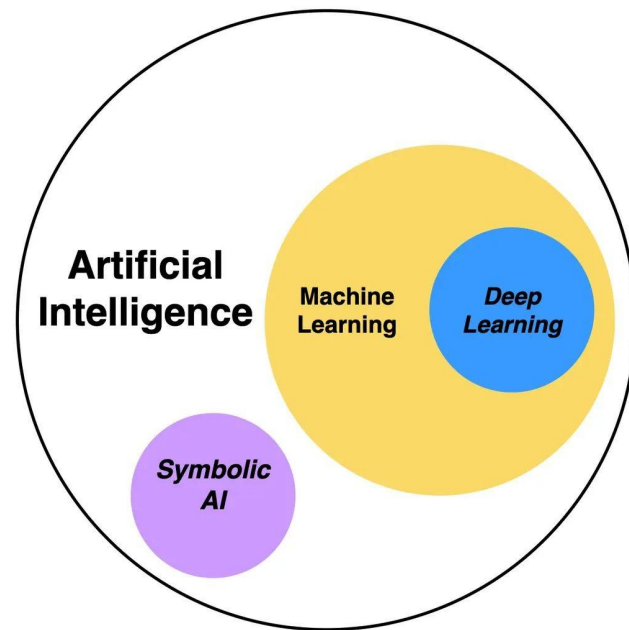
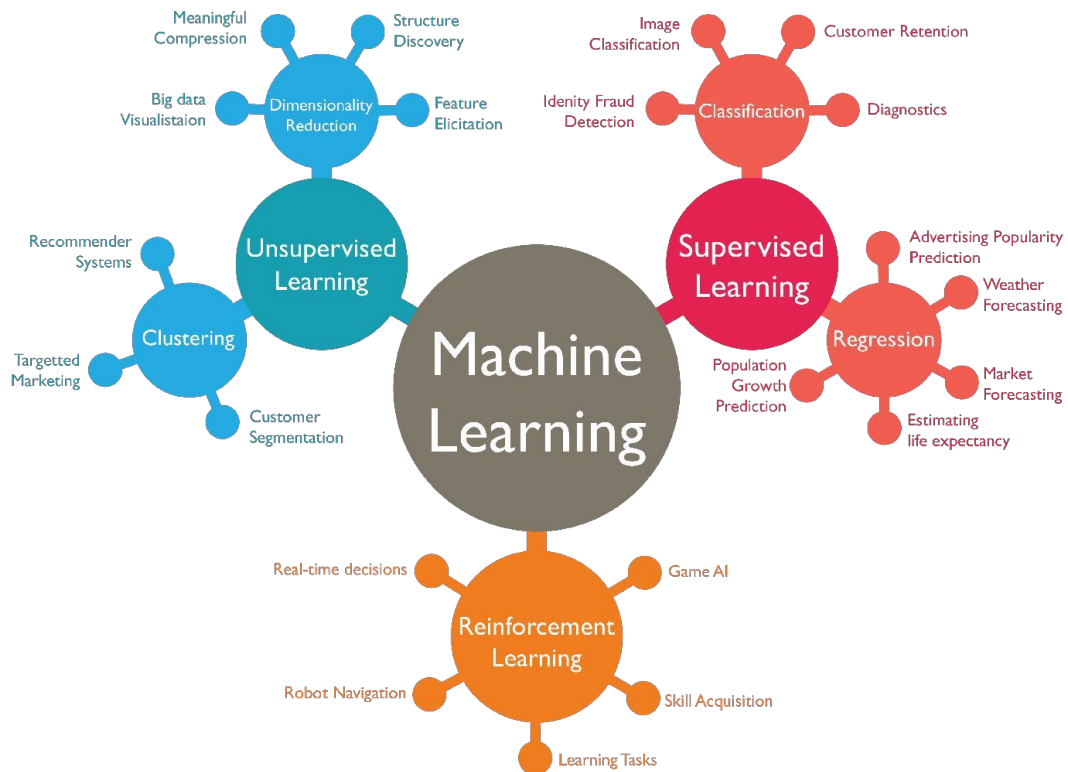


The Trade-off

- Fast enough for serving
- Context can become stale or compressed

Two-tower models separate expensive item computation from real-time user matching.

Machine Learning



Supervised Learning



Student: The right one is a cat!

Teacher: No. This a dog.

Student: Oh... okay, then left one is a dog

also. Teacher: Correct!

- Definition:
Supervised learning is a type of machine learning where a model is trained on labeled data. The algorithm learns to map inputs to outputs based on example input-output pairs.
- Key Components:
 - Input Data (**Features**): Known variables used to predict outcomes.
 - Output Data (**Labels**): Known outcomes or targets the model aims to predict.

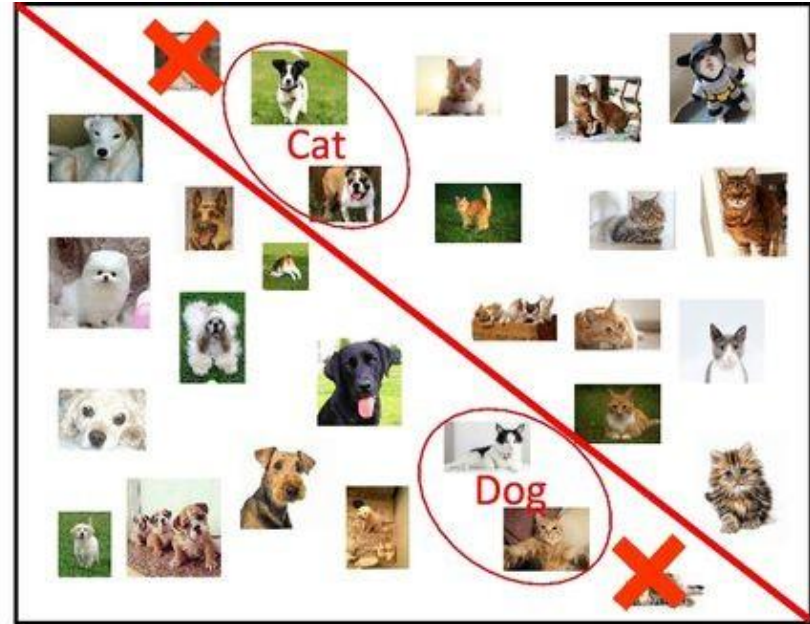
Type of Supervised Learning

Classification (Binary): Only 2 classes

Example: Spam detection



a

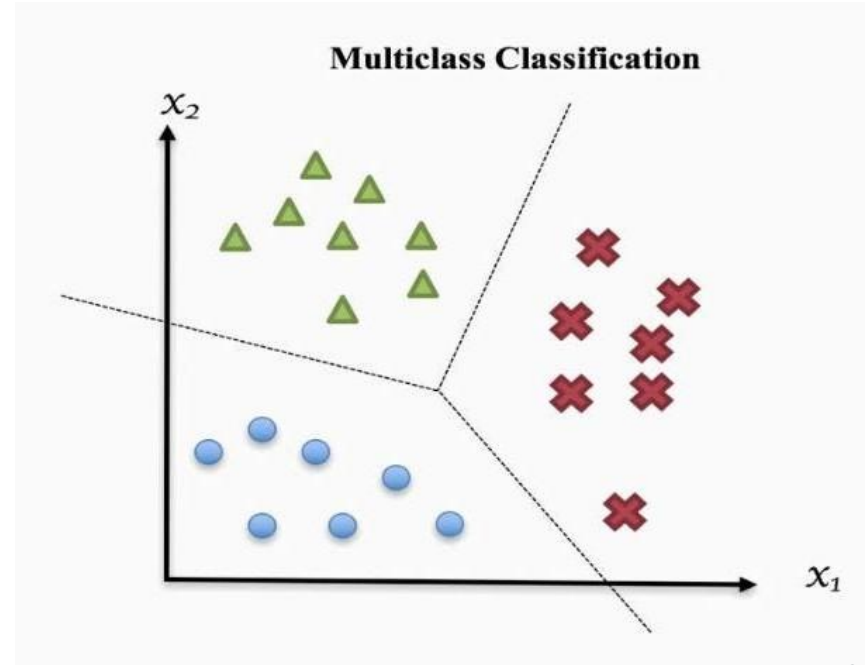
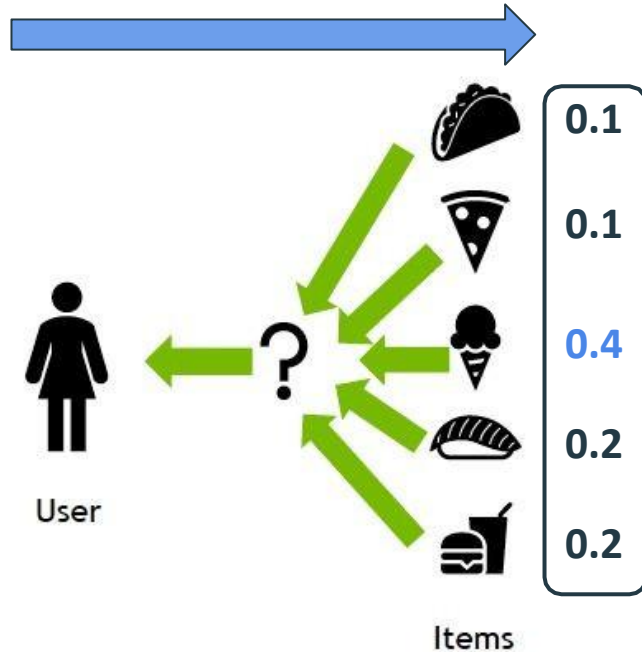


b

Type of Supervised Learning

Classification (Multiclass): More than 2 classes

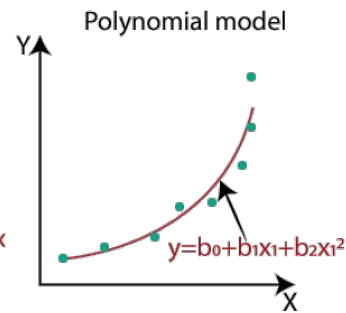
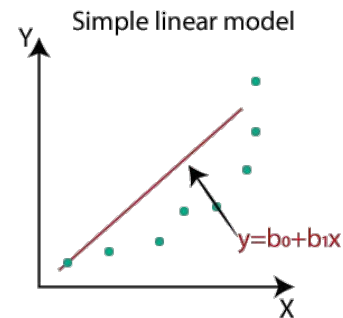
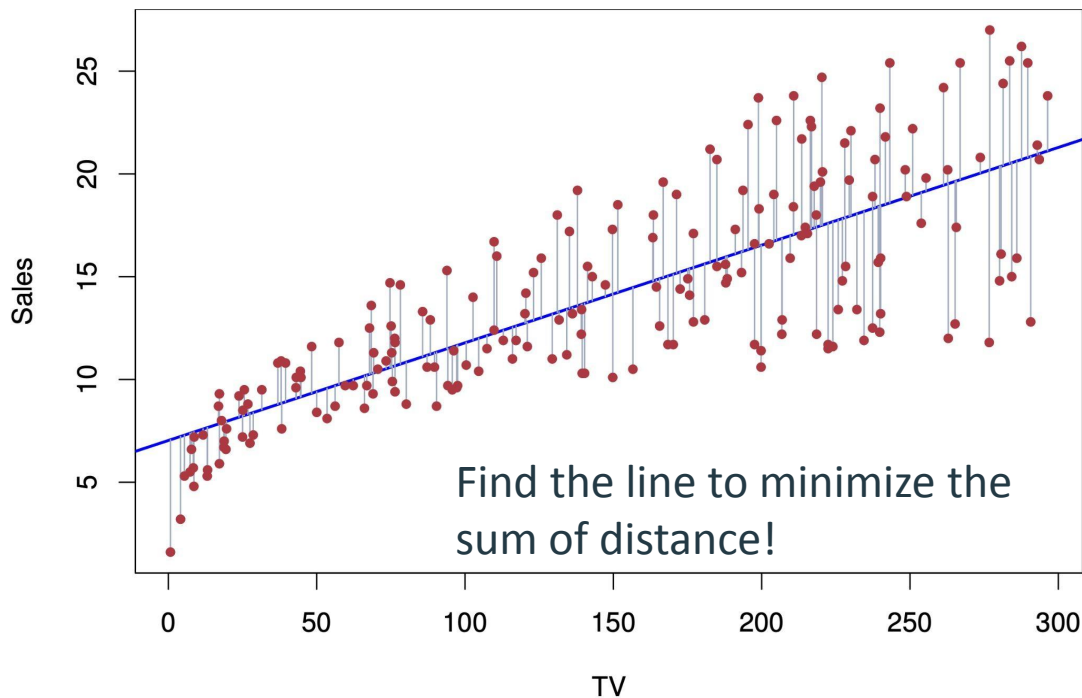
Example: Object recognitions



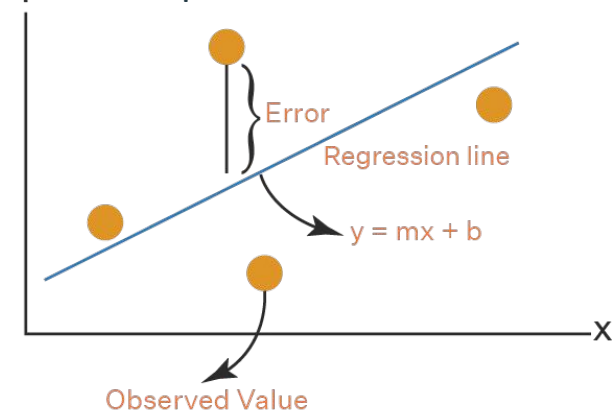
Type of Supervised Learning

Regression: Output is now continuous values

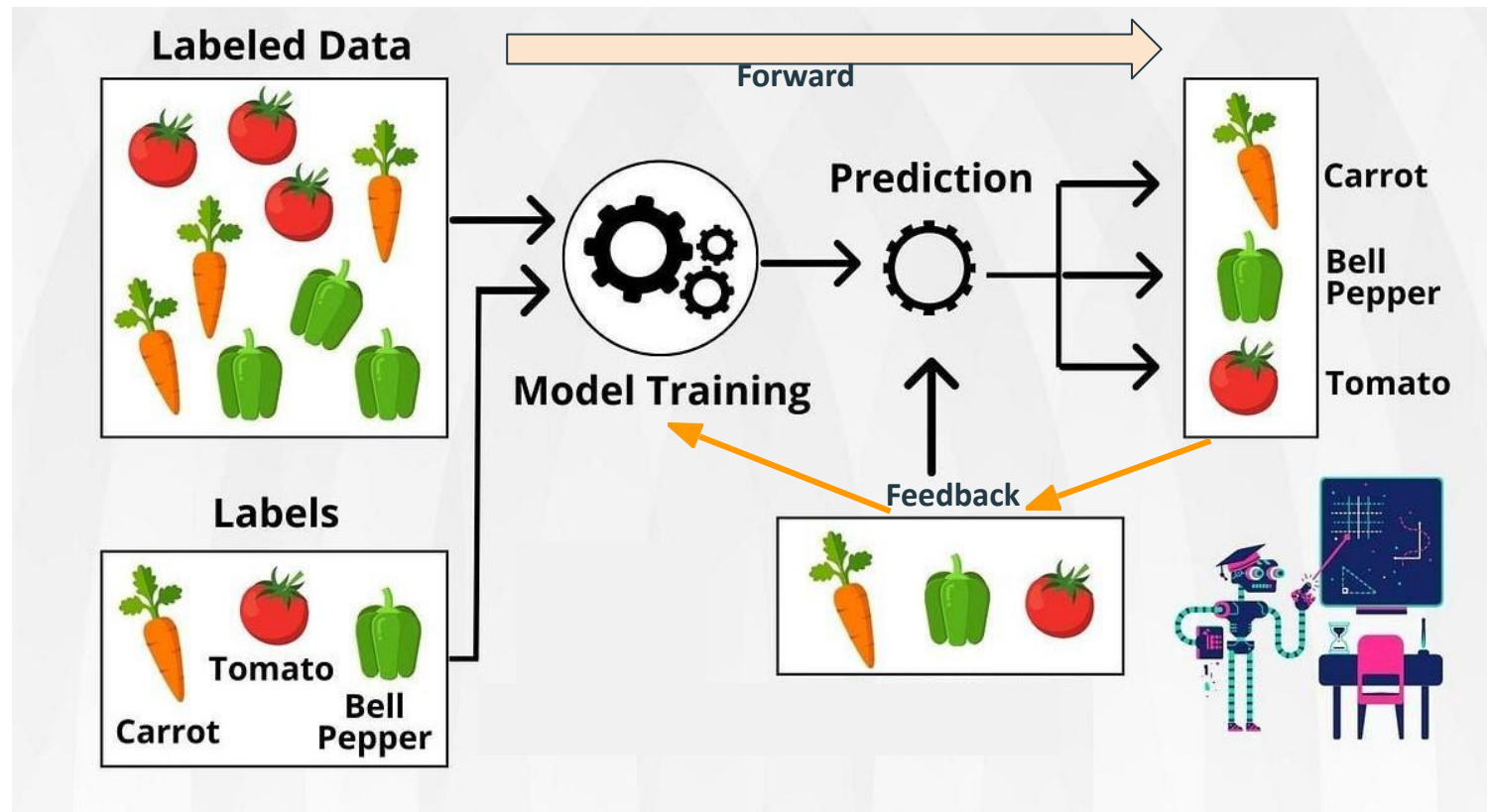
Example: Stock Market Prediction



Least Square Method



Training Supervised Learning



Training Supervised Learning

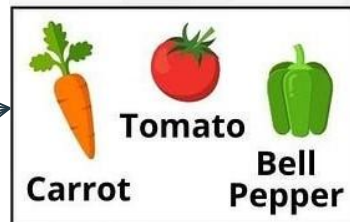
Index	Prob of Carrot, Tomato, and Bell Pepper
1	(0.3 0.2 0.5) => (0.25 0.3 0.45) Update
2	(0.25 0.3 0.45)
3	(0.2 0.4 0.4)
4	(0.15 0.5 0.35)
..	(0.1 0.2 0.7)

Transformation
Formula



Tomato

Labels



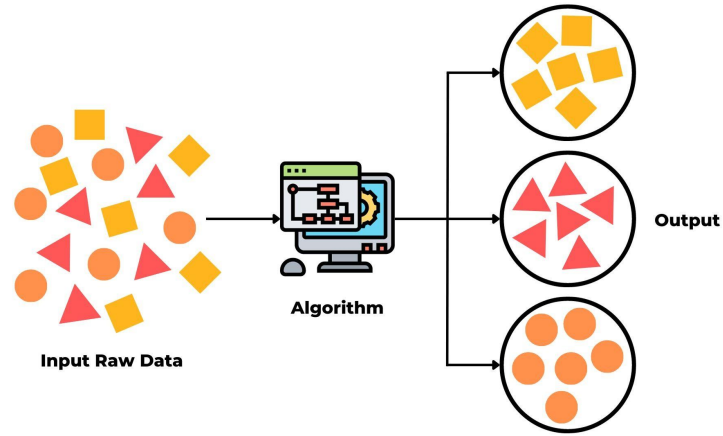
Repeat (Iteration)

	A	B	C	D	E
1	x_0	x_1	x_2	x_3	x_4
2	0.2856096	0.333708	0.8861952	0.3166665	0.388544
3	0.3936127	0.9554749	0.5033009	0.48748	0.0252602
4	0.6678283	0.4833514	0.8402296	0.2364408	0.2593407
5	0.4407239	0.2374104	0.3966326	0.3496683	0.0629024
6	0.2622265	0.2117203	0.1967281	0.5731789	0.5462414
7	0.7209833	0.357229	0.5069117	0.7213588	0.7241538
8	0.1890057	0.1909881	0.0255648	0.7950619	0.8394897
9	0.7827472	0.6916136	0.956046	0.4994256	0.4466799
10	0.5631773	0.8218485	0.0667687	0.9893122	0.7558692
11	0.1593291	0.736723	0.2591398	0.4339377	0.5188055

Unsupervised Learning



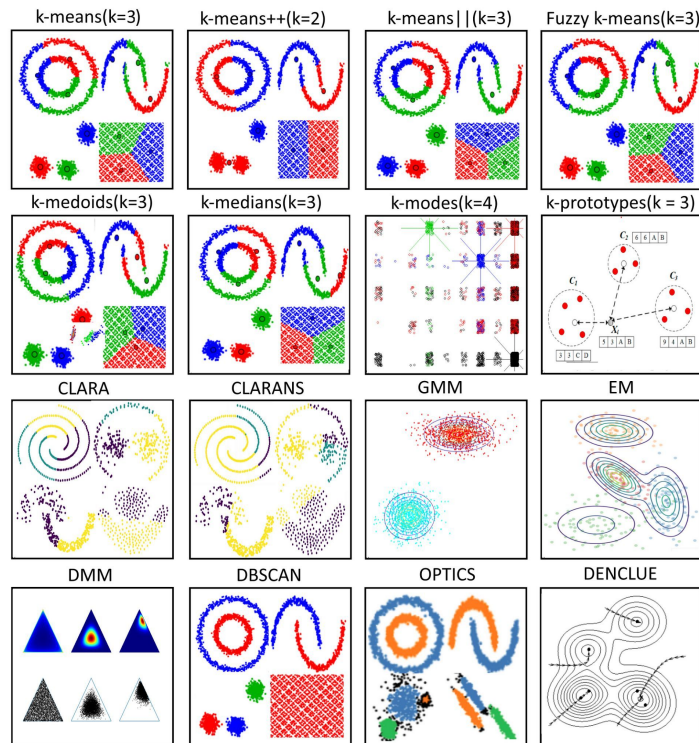
Kid: This and that look similar!



- Definition:
Unsupervised learning is a type of machine learning where the model is trained on data **without labeled responses**. The goal is to identify patterns, structures, or features in the data.
- Key Characteristics:
 - No predefined labels or outputs.
 - Focus on discovering hidden structures.

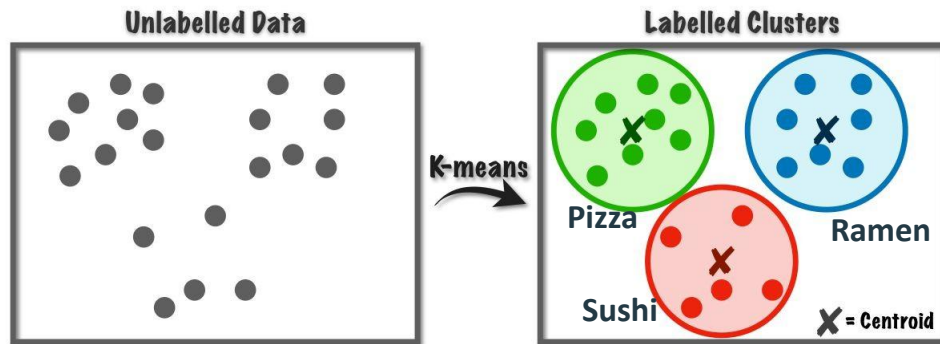
Type of Unsupervised Learning

Clustering: Grouping Similar data based on features



K-Means

Assign data points into the cluster based on the distance of the centroid

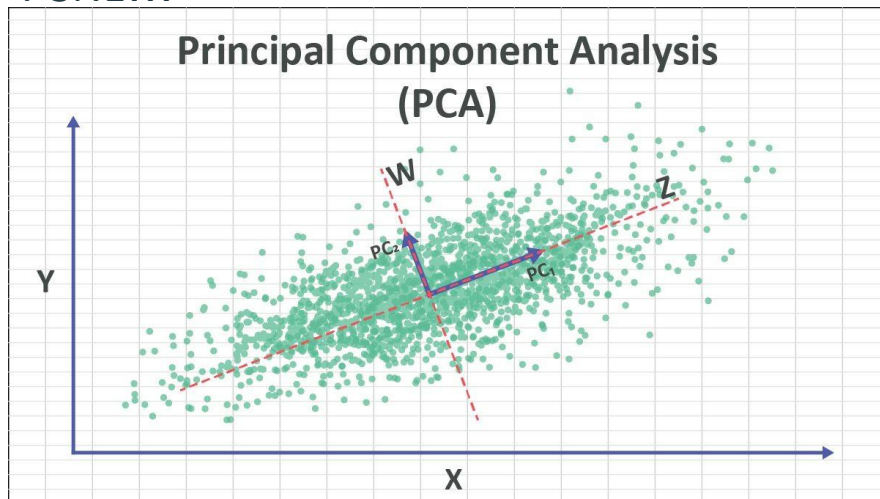


Type of Unsupervised Learning

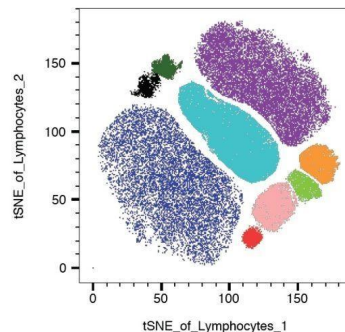
Dimensionality Reduction: Reduce complex features based on similarity or rules

Examples: PCA,

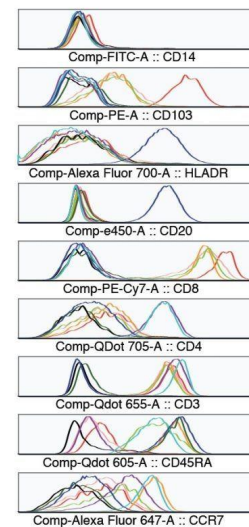
T-SNE...



T-SNE



Sample Name	Subset Name	Count
Tissue Samples_M LN.596613.fcs	9	34331
Tissue Samples_M LN.596613.fcs	8	15646
Tissue Samples_M LN.596613.fcs	7	22942
Tissue Samples_M LN.596613.fcs	6	1183
Tissue Samples_M LN.596613.fcs	5	1359
Tissue Samples_M LN.596613.fcs	4	1427
Tissue Samples_M LN.596613.fcs	3	3050
Tissue Samples_M LN.596613.fcs	2	3821
Tissue Samples_M LN.596613.fcs	1	919



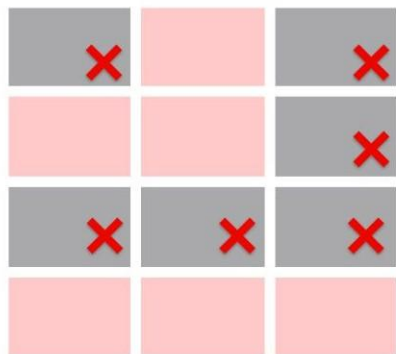
Evaluation Matrix

		predicted condition		
		prediction positive	prediction negative	
true condition	total population			Sensitivity
	condition positive	True Positive (TP)	False Negative (FN) (Type II error)	Recall = $\frac{\sum TP}{\sum \text{condition positive}}$
	condition negative	False Positive (FP) (Type I error)	True Negative (TN)	Specificity = $\frac{\sum TN}{\sum \text{condition negative}}$
		Precision= $\frac{\sum TP}{\sum \text{prediction positive}}$		F1 Score = $\frac{2}{\frac{1}{\text{Recall}} + \frac{1}{\text{Precision}}}$
		Accuracy = $\frac{\sum TP + \sum TN}{\sum \text{total population}}$		

Accuracy: The proportion of total predictions that were correct. It is calculated as the number of correct predictions divided by the total number of predictions.

Evaluation Matrix

Predictive quality metrics



50% correct

Ranking quality metrics



Bad

Good

Behavioral metrics



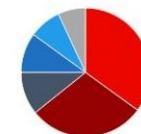
Diversity



Bias



Novelty



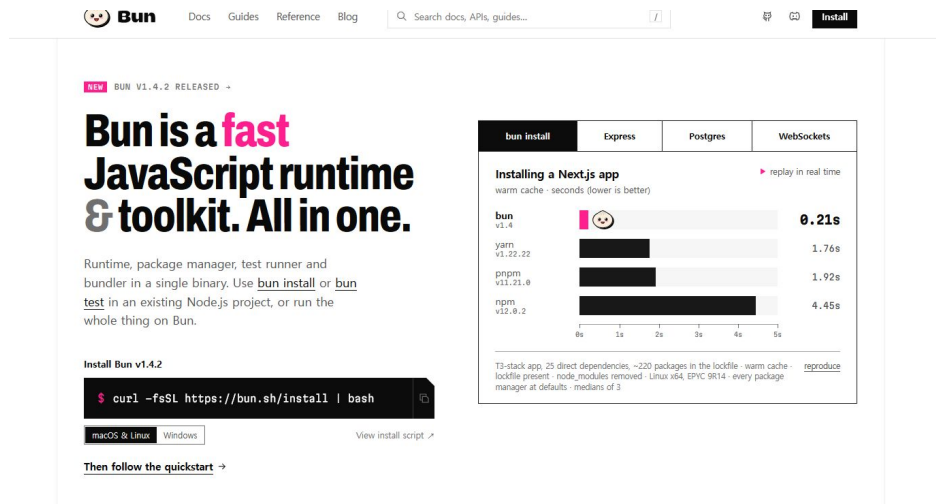
Coverage

Evaluation Matrix Summary

Metric Type	Key Metrics	Purpose
Accuracy	Precision@K, Recall@K, MAP@K, Hit Rate@K	Measure how well recommendations match user preferences (e.g., ratio of relevant items).
Ranking	NDCG@K, MRR	Evaluate ranking quality (e.g., relevance order in top results).
Error	RMSE, MAE	Quantify prediction errors (e.g., difference between predicted and actual ratings).
Other	Diversity, Coverage	Assess variety and breadth of recommendations.

Labs

Install Opencode



Bun Docs Guides Reference Blog Search docs, APIs, guides... 7/ Install

NEW BUN V1.4.2 RELEASED +

Bun is a **fast** JavaScript runtime & toolkit. All in one.

Runtime, package manager, test runner and bundler in a single binary. Use `bun install` or `bun test` in an existing Node.js project, or run the whole thing on Bun.

Install Bun v1.4.2

```
$ curl -fsSL https://bun.sh/install | bash
```

macOS & Linux Windows View install script ↗

Then follow the [quickstart](#) →

Installing a Next.js app ▶ replay in real time
warm cache - seconds (lower is better)

Tool	Time
bun v1.4	0.21s
yarn v1.22.22	1.76s
pnpm v11.11.0	1.92s
npm v10.8.2	4.45s

T3-stack app, 25 direct dependencies, ~220 packages in the lockfile - warm cache - lockfile present - node_modules removed - Linux x64, EPYC 9R14 - every package manager at defaults - medians of 3 [reproduce](#)

Bun:
<https://bun.com/>

Opencode:
<https://opencode.ai/en>

Harness Example:
<https://github.com/dryjins/Controlled-AI/>

MISTRAL AI · DEVSTRAL				
	Devstral 2 123B 60.00% off	128,000	\$0.36 \$0.14	\$1.00 \$0.72
CODING MULTI-GPU				\$0.035 \$0.014
GOOGLE · GEMMA				
	Gemma 4 31B IT FP8 Free - 100% off	262,144	\$0.11 \$0	\$0.35 \$0
LONG CONTEXT FP8				\$0.011 \$0
ZHIPU AI · GLM				
	GLM 5.3 Flash New - 70% off	-	\$0.30 \$0.09	\$1.00 \$0.30
FAST				\$0.009 \$0.009
OPENAI · GPT-OSS				
	GPT-OSS 20B Free - 100% off	20,000	\$0.07 \$0	\$0.25 \$0
DEDICATED				\$0.03 \$0
ORINITH · ORINITH				
	Ornith 1.0 35B	262,000	\$0.14	\$1.00
LONG CONTEXT				\$0.028 \$0
ALIBABA CLOUD · QWEN				
	Qwen3.6 27B Free - 100% discount	262,144	\$0.26 \$0	\$0.00 \$0
CODING LONG CONTEXT				\$0.05 \$0
ALIBABA CLOUD · QWEN				
	Qwen3.6 35B A3B Free - 100% discount	262,000	\$0.12 \$0	\$0.00 \$0
REASONING LONG CONTEXT MOE				\$0.024 \$0

[Models | InferX](#)

What Are OpenCode and an AI Harness?

OpenCode

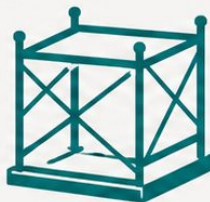


**An open-source
AI coding environment**

- Reads your repository
- Follows project instructions
- Helps create, inspect, and revise code



AI Harness



**A structured environment
that makes AI work
controlled and repeatable**

- Rules and roles
- Skills and tools
- Inputs and outputs
- Verification checks



Course Workflow

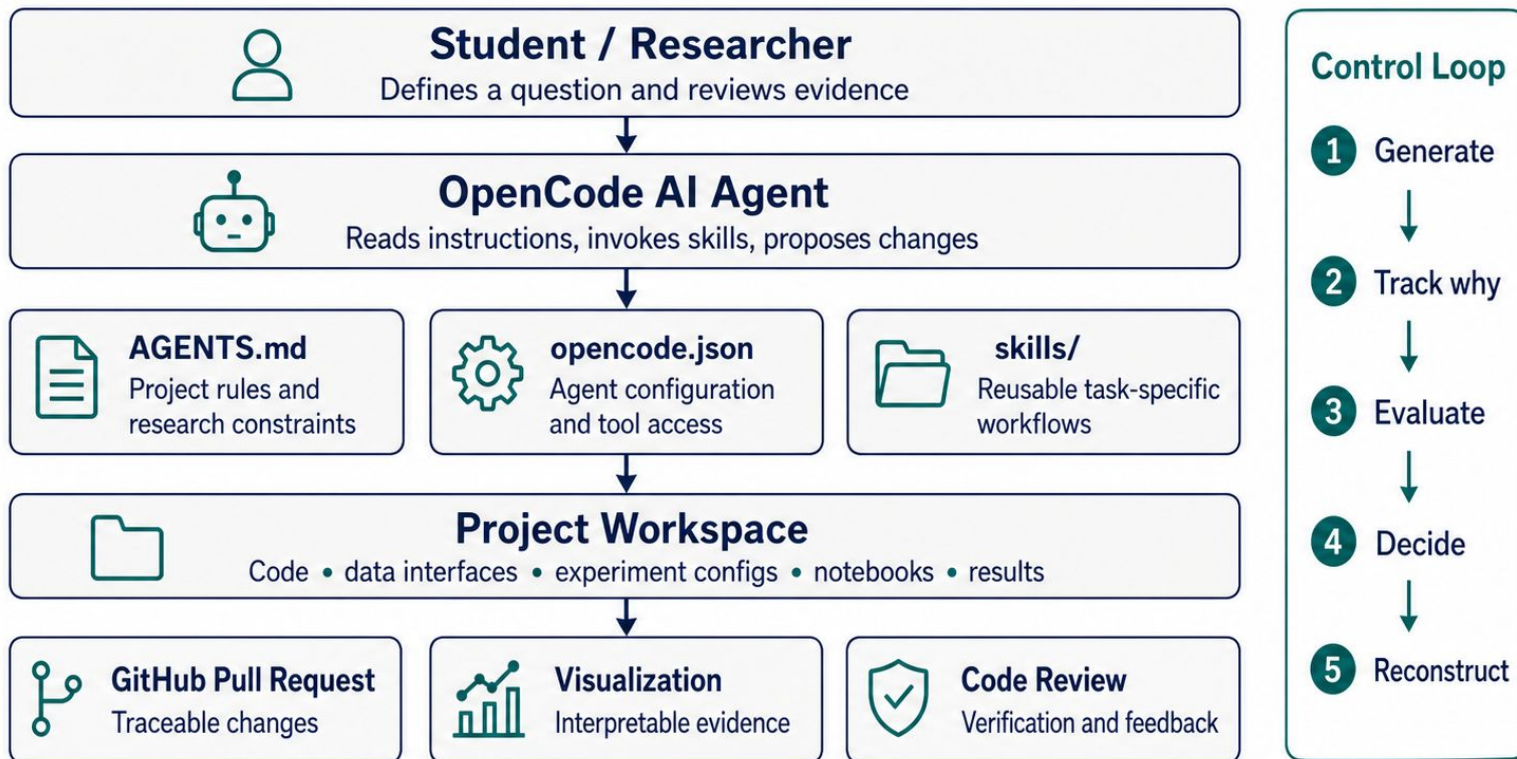


**Question → Code →
Experiment → Evidence →
Review**

- GitHub records the work
- Students verify AI suggestions
- Results support a claim

The harness does not replace thinking. It makes AI-assisted work reproducible, inspectable, and learnable.

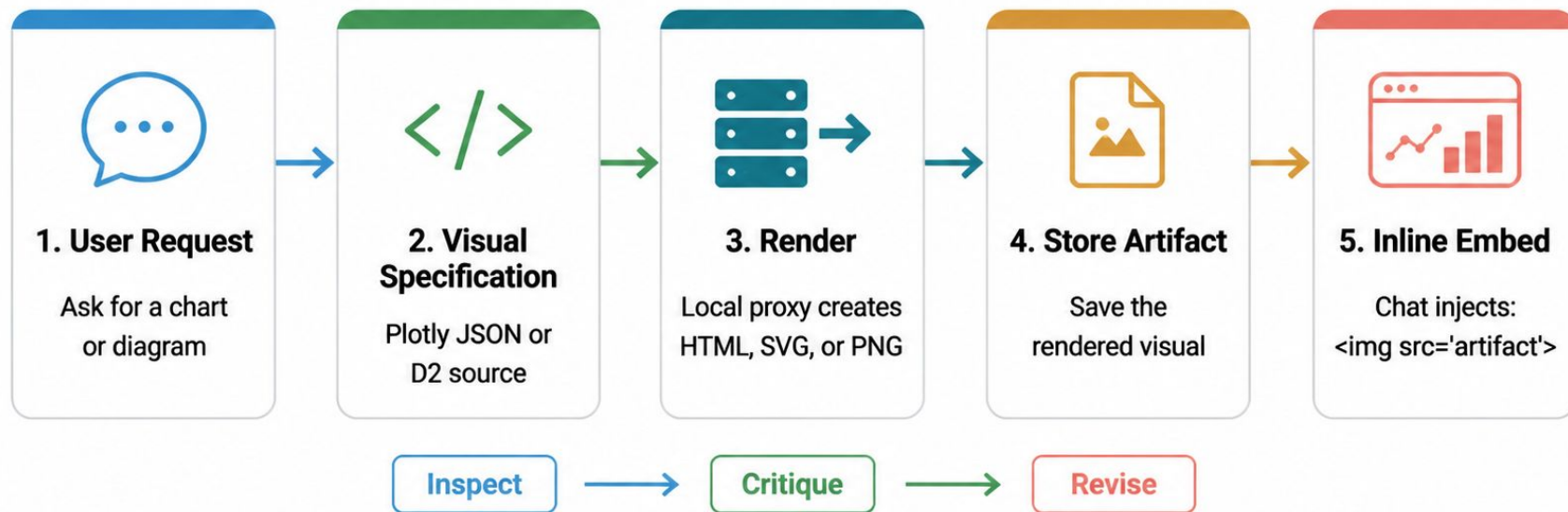
OpenCode Harness Architecture



A harness turns AI assistance into a controlled, reproducible learning workflow.

How OpenCode Embeds Visuals in Chat

Plotly or D2 output is rendered into an image artifact, then inserted inline with an HTML `<img>` tag.



The agent generates a specification; the renderer produces the image; the chat displays the image.

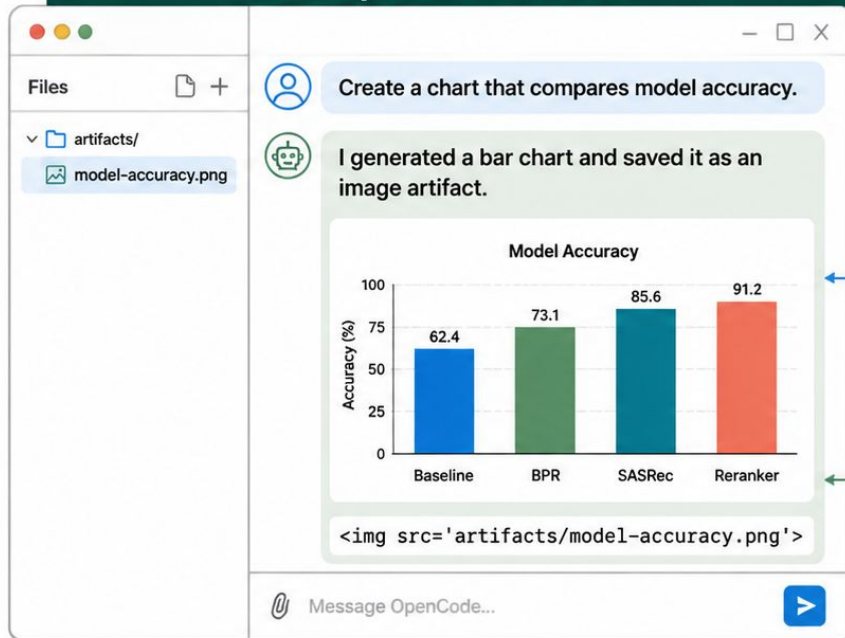
Show and Explain Your Work

Bring one visualization-related skill or feature you prepared in advance.

Your 3–5 Minute Demo

- 1  Show your OpenCode chat or project files
- 2  Demonstrate one working example
- 3  Show the final visual output inside the chat
- 4  Explain what you built and how it works
- 5  Describe one challenge or next improvement

Example: What to Show



The screenshot shows a chat window with a file explorer on the left containing a folder 'artifacts/' and a file 'model-accuracy.png'. The chat history includes a user message 'Create a chart that compares model accuracy.' and an AI response 'I generated a bar chart and saved it as an image artifact.' Below the text is a bar chart titled 'Model Accuracy' showing accuracy percentages for four models: Baseline (62.4%), BPR (73.1%), SASRec (85.6%), and Reranker (91.2%). Below the chart is an image tag `<img src='artifacts/model-accuracy.png'>`. At the bottom is a text input field 'Message OpenCode...' and a send button.

Model	Accuracy (%)
Baseline	62.4
BPR	73.1
SASRec	85.6
Reranker	91.2

1. Generated artifact

2. Visible inside chat

You do not need a perfect system. Show what you built and explain your design choices.

Homework: Random Lunch Generator

Overview:

Simple web app recommends random lunch menus (e.g., Pizza, Sushi).

Goal: Learn basics of web dev & recommendation in 10 mins.

Tech: HTML (structure), CSS (design), JS (random logic), GitHub Pages (deploy).

How It Works:

1. Click "Generate Lunch!" button.
2. JS picks random item from list using `Math.random()`.
3. Display: "Today's pick: Sushi! 🍣"

Key Insight: Starts random – evolve to AI-personalized later!

TODO: Try changing the current prompt and code to understand and fix the issue of images sometimes not displaying.

[RecSys-LLMs/week1/prompt.md at main · dryjiins/RecSys-LLMs](#)

[Random Lunch Menu Generator](#)

